1 Naive Bayes Classification

[1]:
```
# importing libraries and the dataset
import pandas as pd

df = pd.read_csv('play_tennis.csv')
df
```

[1]:

	outlook	temp	humidity	wind	play
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

[2]:
```
# counting Yes and No values
class_counts = df['play'].value_counts()
class_counts
```

[2]:
```
play
Yes    9
No     5
Name: count, dtype: int64
```

[3]:
```
# counting attribute values for each class
for column in df.columns[:-1]:
    print("\n", column)
    print(pd.crosstab(df[column], df['play']))
```

```
 outlook
play      No  Yes
outlook
Overcast   0    4
Rain       2    3
Sunny      3    2

 temp
play  No  Yes
temp
Cool   1    3
Hot    2    2
Mild   2    4

 humidity
```

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```
play      No  Yes
humidity
High       4    3
Normal     1    6

 wind
play    No  Yes
wind
Strong   3    3
Weak     2    6
```

[4]:
```
# new instance
X = {'outlook': 'Sunny', 'temp': 'Cool', 'humidity': 'High', 'wind': 'Strong'}

# conditional probabilities
for cls in class_counts.index:
    print("\nProbabilities for play =", cls)

    for attribute, value in X.items():
        count = len(df[(df[attribute] == value) & (df['play'] == cls)])
        total = class_counts[cls]
        probability = count / total
        print(attribute, "=", value, ":", count, "/", total, "=", round(probability,␣
 ↪4))
```

```
Probabilities for play = Yes
outlook = Sunny : 2 / 9 = 0.2222
temp = Cool : 3 / 9 = 0.3333
humidity = High : 3 / 9 = 0.3333
wind = Strong : 3 / 9 = 0.3333

Probabilities for play = No
outlook = Sunny : 3 / 5 = 0.6
temp = Cool : 1 / 5 = 0.2
humidity = High : 4 / 5 = 0.8
wind = Strong : 3 / 5 = 0.6
```

[5]:
```
# calculating Naive Bayes probabilities
probabilities = {}

for cls in class_counts.index:
    # prior probability
    probability = class_counts[cls] / len(df)

    # multiply by conditional probabilities
    for attribute, value in X.items():
        count = len(df[(df[attribute] == value) & (df['play'] == cls)])
        total = class_counts[cls]
        probability *= count / total

    probabilities[cls] = probability
    print("Probability for", cls, "=", probability)
```

```
Probability for Yes = 0.005291005291005291
```

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```
Probability for No = 0.02057142857142857
```

[6]:
```
# final prediction
prediction = max(probabilities, key=probabilities.get)
print("Final Prediction:", prediction)
```

```
Final Prediction: No
```

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